

10 A hot water tank of heat capacity 6500 J K^{-1} contains 14 kg of water at 30°C .

The specific heat capacity of water is $4200 \text{ J kg}^{-1}\text{K}^{-1}$ and specific latent heat of vaporisation of water is $2.26 \times 10^6 \text{ J kg}^{-1}$.

What is the time taken to convert all the water to steam using a heater coil of power 5.2 kW ?

- A** 800 s
- B** 6000 s
- C** 6200 s
- D** 7000 s

11 A fixed mass of an ideal gas is heated at constant volume